Quantifying Look-Ahead Bias via Causal, Volatility-Aware Motif Tokenization in Financial Time Series

DA4022 Course Project Proposal · Team of 5

Motivation. BPE motif tokenizers (Götz et al., 2026) are validated zero-shot on held-out datasets. Used for walk-forward forecasting on a single market, they can leak future data in two ways. *Fit-time:* normalization statistics, the merge table or the decoding table are estimated on data that overlaps the test period. *Encode-time:* rank-ordered BPE is not incremental (Zhang et al., 2026). A stream tokenized once and then split at the forecast origin carries post-origin information in its token boundaries. Tokenizing the context alone instead causes partial-token bias (Phan et al., 2024).

Method. We use 5-minute Binance crypto returns (2019–2025), normalized by trailing EWMA volatility. Hourly scale tokens carry absolute volatility and serve as hard token boundaries (UniTok). Tokenization is Götz-style BPE with fixed bins and a fixed merge count, so encoding is incremental at every hour-aligned forecast origin. A causal RoV×SEI gate inspired by PATK (Liao et al., 2026) shortens tokens in volatile hours. A ~1M-parameter GPT with conditional decoding forecasts next-hour and next-day realized volatility.

Experiments. Starting from this causal reference, we switch on one leakage channel at a time: per-window normalization, merge table, decoding table, token boundaries and a non-causal FFT. Each leak is applied in both training and testing. As a negative control, we simulate a GARCH process whose true (oracle) volatility forecast is known, so no leak-free model can beat it. As a positive control, we use a deliberately gross normalization leak. We use six expanding walk-forward folds, plus 2025 held out and scored only once at the end. Evaluation uses QLIKE with block-bootstrap confidence intervals, Holm-corrected superiority tests and TOST equivalence tests, plus Diebold–Mariano and Giacomini–White comparisons against HAR-RV and GARCH(1,1).

Novelty. We audit, channel by channel, the leakage inside time-series tokenizers; prior work studies dataset-level or windowing leakage. To our knowledge, this is the first measurement of the trade-off between token-boundary leakage and partial-token bias for time-series tokenizers. We also contribute a causal, PATK-inspired gate on token length. Deliverables are a per-channel inflation table, reproducible code, the report and the video.